

Referee report — "ContEst: Control and Estimation Co-Design ..."

Recommendation: **Major revision**

The paper makes a genuinely useful observation — that for the *value* of a convex inner program the design gradient is available for free from the dual the solver already returns, and that this unifies the LQG/H ∞ /eKF-MPC co-design variants under one envelope-theorem identity. The framing (bringing the estimator/sensing axis into control co-design) is well-motivated and the writing is mostly clear. However, the central theorem's proof is only a sketch, one newly-featured exactness result is asserted rather than proved, and the numerical section has methodological confounds — including one example that contradicts the paper's own stated evaluation protocol. These are fixable, but they are substantive.

Major comments — mathematics and proofs

M1. The proof of Theorem 1 (envelope gradient) is a heuristic, not a proof. The proof (l.491–502) differentiates $V(\theta) = L(z^*(\theta), \mu^*(\theta), S^*(\theta), \theta)$ via the chain rule and asserts the terms in $\partial_\theta z^*$ and $\partial_\theta(\mu^*, S^*)$ vanish. Two problems:

- It invokes **differentiability of the solution map** $\theta \mapsto (z^*, \mu^*, S^*)$. That is *stronger* than what you need and, more awkwardly, it is exactly the object your own selling point (l.533–540, "no implicit differentiation") says you avoid — establishing C^1 -ness of the solution map is normally done through the IFT on the KKT system. This reads as internally inconsistent.
- For the **conic** term, $\nabla_S L = G(z^*, \theta) \in \mathcal{K}$ is *not* zero; the claimed cancellation of $\langle \partial_\theta S^*, G(z^*, \theta) \rangle$ is the crux and is precisely what needs strict complementarity + the perturbation theory you cite. The one-line " $= \partial_\theta L$ " glosses it.

Fix: either invoke Bonnans–Shapiro (Thm 4.24-type) directly and drop the chain-rule hand-wave, or give the clean Danskin/saddle argument $V(\theta) = \min_z \max_{S \in \mathcal{K}^*} L$ with a supporting-minorant identification of the gradient. The latter needs only the *unique* primal–dual, not a differentiable solution map.

M2. Corollary "H ∞ fixed- γ design: exact gradient" rests on an unproven regularity claim. You now (correctly) restructure the H ∞ section around the fixed- γ SDP being the exact case, but the corollary just states "[eqrefeq](#) : *Hinf*_{fixed} is strictly feasible with a nondegenerate, strictly complementary dual, so Assumption 1 holds." There is no proof that the fixed- γ BRL SDP has a unique/nondegenerate dual — in contrast to the H $_2$ case, where Appendix B proves dual uniqueness constructively ($S_{11}^* = P$). Since the ADCS control gradient depends on this, it needs support. The cleanest route is already in your hands: the fixed- γ GARE has a stabilizing solution $X(\theta)$ that is locally C^1 for γ above the optimal attenuation (Remark "Scalable game-Riccati gradient"), so $V_{wc} = \text{tr}(XW_\theta)$ is differentiable; then argue the SDP dual recovers that gradient. As written, the exactness is asserted.

M3. Appendix C (subgradient existence) proves the right thing but by a detour. Your weak-duality bound gives a smooth *minorant* $g(\theta') = \min_z L(z, \lambda^*, \theta')$ with $g \leq V$ and $g(\theta) = V(\theta)$; that alone yields $\partial_\theta L(z^*, \lambda^*, \theta) = \nabla g(\theta) \in \partial^- V(\theta) \subseteq \partial_{\text{Clarke}} V(\theta)$ directly — cleaner, and it *identifies the returned dual as the subgradient*, which is what you want. The current "Lipschitz $\Rightarrow$ Rademacher $\Rightarrow$ pass

to the limit" ending (l.1601–1608) actually needs outer-semicontinuity of the dual map to conclude the *specific returned* dual is a limit of nearby gradients, which you do not establish. Recommend recasting Prop. 2's proof around the minorant.

M4. Two of the four regularity assumptions are conflated in the abstract/intro. Line 142 says the gradient is exact "under a unique inner minimizer"; exactness needs *both* a unique minimizer (A2) *and* a unique dual / strict complementarity (A3), as your contribution bullet (l.148) and Assumption 1 correctly state. Align the intro.

M5. Info-state formulation vs. state feedback (l.275). The infinite-horizon inner problem is written $\min_{\kappa} \lim_k \mathbb{E} \ell(x_k, \kappa(x_k))$ — full-state feedback — which contradicts the paper's thesis that the control is a function of the *information state*, not x_k . Should be $\kappa(\mathcal{Z}_k)$ / a function of the filtering density. Same care in eq. (mpc): the linearized information-state dynamics are *affine* (a constant offset enters the covariance recursion via $+W$); the offset (and its θ -dependence if $W = W_{\theta}$) is dropped from eq:gradmpc without comment.

Major comments — numerical studies

N1. The eKF-MPC baseline is confounded (PLL; and the eKF-ADCS variant). For the nonlinear studies the *baseline* runs a **certainty-equivalence controller** (eKF mean only) at θ_{nom} , while the *optimum* runs the **information-state MPC** at θ^* ([example_pll.jl:147–153](#), [example_adcs.jl:128–131](#); stated in passing at l.1371). So the reported reduction mixes two changes: the design move $\theta_{\text{nom}} \rightarrow \theta^*$ **and** a controller-class upgrade CE→info-state. To attribute the gain to *co-design*, the baseline must hold the controller fixed (info-state MPC at θ_{nom}). As reported, the numbers overstate the design contribution and are not an apples-to-apples design comparison.

N2. The distillation study contradicts the paper's stated protocol. The paper says (l.1004–1006, 1028–1033) that for the nonlinear studies it reports **Monte-Carlo realized closed-loop** cost "to confirm each optimized design improves the true cost, not merely the surrogate." But [example_distillation.jl](#) (l.171–186) **skips MC entirely** ("MC sampling skipped for this study") and reports the **open-loop MPC surrogate value** J_c plus a design-time covariance rollout, with J_{det} *defined as the residual* $J_c - J_{\text{est}}$. Thus the distillation reductions (22.8% / 1.9% / 6.5%) are exactly the surrogate/open-loop numbers the paper claims not to rely on. Either run the MC or correct the methodology text; as it stands this is a direct paper-vs-code contradiction, and the 1.9% control figure in particular is unverified against the true cost.

N3. No comparison against a competent sequential design. The paper's structural argument is that co-design captures an interior trade "a control-only design cannot represent" (l.1122–1126). But every reduction is measured against a naive/uniform θ_{nom} , and you honestly caveat (l.1374–1380) that the baselines are "plausible rather than carefully hand-tuned." The decisive co-design experiment is *joint* vs. *sequential*: place sensors first by a standard criterion (A-optimal / greedy submodular) and design control second, then show the joint optimum beats that pipeline. Without it, the experiments show only that "optimizing beats not optimizing," not that *jointness* pays — which is the paper's actual claim. This is the single most important addition, and it is directly on point for an Automatica co-design paper.

N4. Monte-Carlo reporting is incomplete. The sample count M , horizon T , and (critically) **confidence intervals** are not given in the text; a single seed (20240624) is used. Please report M , standard errors/CIs, and preferably use **common random numbers** (paired baseline/optimum) for a variance-

reduced comparison. Absent CIs, small effects (distillation control 1.9%) may be within Monte-Carlo noise.

N5. Public artifact does not match the manuscript. The repository's "run all paper examples" driver runs a **four-study** set — `example_adcs.jl` (eKF-MPC) and `example_mtdc.jl` (HVDC) — whereas the paper describes **three** studies with ADCS as the $H_\infty+H_2$ SDP study (`example_adcs_hinf.jl`). The reproducibility artifact should reproduce the manuscript's studies.

Minor / consistency

- **"Four" vs "three" studies:** contribution bullet (l.153) claims "four realistic co-design studies"; the body, Table 6, and summary all say three. (Consistent with N5 — a leftover from when MTDC was included.)
- **ADCS arithmetic (l.1119–1120):** J_c is reported as -31.9% and J_{tot} as -31.7% , both from an identical $0.224 \rightarrow 0.153$; they must be equal (31.7%). Also the displayed 3-digit costs don't quite reproduce the headline percentages ($0.150 \rightarrow 0.091$ is 39.3% , not 39.5% ; $0.074 \rightarrow 0.062$ is 16.2% , not 16.4%) — give one more digit or reconcile.
- **Notation:** eq. (mpc) uses B_θ but eq:gradmpc uses $\mathcal{B}_\theta$ — unify. `$J_{\mathrm{sc}}^{\star}$` brace typo (l.864). Typos: "accuratel" (l.294), "similarly", "the the".
- **H_∞ filter "exact transpose dual"** (l.640–648): fine for what you do, but flag that you only combine *state-feedback* H_∞ with H_2 estimation — H_∞ *output-feedback* is not separable, and the current wording could mislead.
- **Assumption A4** is called "a modeling convenience," but it is load-bearing for the $O(r_x^2 r_\theta)$ cost claim; say so.

What is solid

The H_2 /LQG regularity proof (Appendix B) is the strongest part — the constructive dual-uniqueness argument giving $S_{11}^* = P$ and $S_{11}'^* = \Sigma^\varepsilon$ is correct and elegant, and it rightly grounds the "same object" claim linking the LMI-dual and Riccati gradients. The complexity analysis ($O(r_x^6)$ IP scaling, with the $2.6\text{s} \rightarrow 30\text{s}$ timing matching $(30/20)^6$) is convincing, and the honesty of Remark "Where exactness stops" about the eKF-MPC surrogate is commendable — the issue in N2 is that the distillation *code* doesn't live up to it.